How to Identify Relationships in an ERD

INF10025 Data Management and Analytics - study guidance, with the Pass 4j (SwinEats) worked example

1. Where relationships come from

Once the entities are drawn, relationships are found by reading the narrative again and asking structured questions about every pair of entities. Follow the steps below in order; each step feeds the next.

2. Six steps to identify a relationship

Step	What to do	Question to ask
1	Find the verbs that connect two entities in the narrative.	Which two nouns does this verb link? (e.g. drivers **do** jobs)
2	Decide the cardinality in each direction.	Can ONE A be linked to many B? Can ONE B be linked to many A?
3	Decide the optionality at each end.	Must every A have a B, or may it have none? (double bar vs circle)
4	Look for attributes that belong to the link itself.	Does this fact (date, fee, grade) describe A alone, B alone, or the pairing of A and B?
5	Expand every many-to-many (M:M).	Replace the M:M with a weak (intersection) entity and two 1:M relationships.
6	Name and mark the relationships.	Give each a short verb name; underline the names of the relationships that supply the weak entity identifier.

Reading the answers from step 2: one-to-many (1:M) if only one direction is "many"; many-to-many (M:M) if both directions are "many". A one-to-one answer is rare in this unit.

3. Notation used in the lectures

Symbol / style	Meaning
Rectangle	Strong entity
Double outline rectangle	Weak (intersection) entity - part of its identifier is borrowed from the entities it connects
Circle on a stem	Attribute; an underlined name is the identifier (or the partial identifier of a weak entity)
Double bar on a line end	Exactly one at that end of the relationship
Circle plus crow's foot	Zero or many at that end of the relationship
Underlined relationship name	Identifying relationship: the many-side entity borrows the identifier of the one-side entity

4. Worked example - Pass 4j (SwinEats)

Narrative (key facts): jobs have an ID, title, duration and description. Drivers have a name, address and mobile phone. Some drivers do more than one job, and the same job can be done by multiple drivers. When a driver does a job, the date and a fixed fee are recorded.

Step	Result for SwinEats
1. Verbs	Drivers **do** jobs: one relationship between DRIVER and JOB.
2. Cardinality	One driver can do many jobs, and one job can be done by many drivers, so the relationship is **M:M**.
3. Optionality	A driver may not have done any job yet, and a job may never have been done: zero or many at the assignment end.
4. Link attributes	DoDate and Fee describe the pairing of a driver with a job, not the driver or the job alone.
5. Expand M:M	Create the weak entity ASSIGNMENT5520 with two 1:M relationships: DRIVER5520 1:M ASSIGNMENT5520 and JOB5520 1:M ASSIGNMENT5520.
6. Name / mark	"does" and "is done in", both underlined, so ASSIGNMENT5520 borrows MobileNo and JobID and adds its own DoDate.

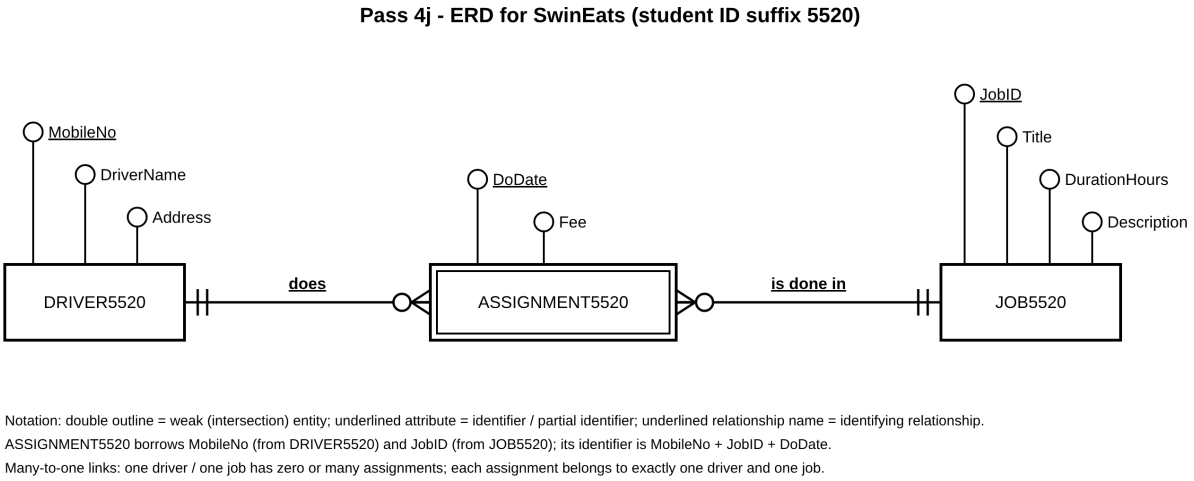


Figure 1: ERD for Pass 4j in the lecture notation.

5. From relationships to the relational schema

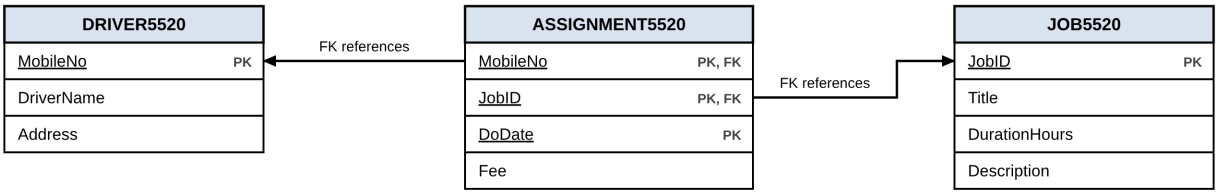
Rule	Applied to SwinEats
Every entity becomes a table; its identifier becomes the primary key.	DRIVER5520 (MobileNo), JOB5520 (JobID)
In a 1:M relationship the foreign key goes in the table on the MANY side.	MobileNo and JobID are foreign keys in ASSIGNMENT5520
A weak entity's primary key = the borrowed identifiers + its own partial identifier.	Primary key (MobileNo, JobID, DoDate)
No surrogate keys: use natural identifiers from the narrative.	MobileNo is used instead of an invented DriverID

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DRIVER5520 (MobileNo, DriverName, Address)
Primary Key (MobileNo)

JOB5520 (JobID, Title, DurationHours, Description)
Primary Key (JobID)

ASSIGNMENT5520 (MobileNo, JobID, DoDate, Fee)
Primary Key (MobileNo, JobID, DoDate)
Foreign Key (MobileNo) references DRIVER5520
Foreign Key (JobID) references JOB5520
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Pass 4j - Relational Schema (ERM) for SwinEats (student ID suffix 5520)



DRIVER5520 (MobileNo, DriverName, Address)
JOB5520 (JobID, Title, DurationHours, Description)
ASSIGNMENT5520 (MobileNo, JobID, DoDate, Fee) Foreign Key (MobileNo) references DRIVER5520; Foreign Key (JobID) references JOB5520

Key: underlined = primary key (ASSIGNMENT5520 has a composite primary key); PK = primary key; FK = foreign key; arrow points from the foreign key to the primary key it references.

Figure 2: Relational schema (ERM) for Pass 4j.

Why DoDate alone is not a primary key

A primary key must be unique for every row. Two different drivers can do two different jobs on the same day, so DoDate alone would repeat. The combination of who (MobileNo), which job (JobID) and when (DoDate) identifies exactly one assignment.

6. Checklist before you submit

- Every entity name ends with the last 4 digits of your student ID (e.g. DRIVER5520).
- Every M:M has been expanded into a weak entity and two 1:M relationships.
- Relationship names are written and the identifying ones are underlined.
- The weak entity has a double outline and its many-side links point at it.
- The schema has no surrogate keys, and every foreign key is also listed as a column of its table.
- Two rows of test data exist for each entity.